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Integrating Clinical and Socioeconomic Features in an Explainable AI Model for Paediatric Respiratory Disease Prediction

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***Abstract*—As a major reason for hospitalization and death, paediatric respiratory diseases necessitate precise analytical assistance systems. The explainable machine learning architecture presented in this study was developed using 801 paediatric patients' medical and socioeconomic information. Random Forest-based feature selection, Min-Max scaling, and random oversampling were used as preprocessing techniques to improve feature significance while addressing disparity. Several ensemble techniques and six classifiers were examined; Random Forest performed best (99.56% accuracy, AUC = 1.00). Applying SHAP and LIME to assure accessibility highlighted key indicators like oxygen saturation, inflammatory biomarkers, and intensive care unit stay. Better paediatric respiratory care results, early danger detection, and clinician credibility are made possible by the suggested approach's combination of powerful forecasting abilities and clear justifications.**

Keywords—Pediatric respiratory disease prediction, Machine learning, Random Forest, Gradient Boosting, Ensemble learning, SHAP, LIME

I. Introduction

This Prolonged breathing issues such as asthma, bronchospasm, laryngeal dyspnea, bronchiolitis, and staphylococcal pneumonia pose a serious risk to public health and are the fourth leading cause of death [1]. The prevalence of such illnesses is higher in children. Respiratory illness-related pediatric intensive care unit (PICU) mortality has far-reaching consequences for both children's well-being and the healthcare system as a whole [2]. Paediatric breathing disorders such as severe pneumonia, acute respiratory distress syndrome (ARDS), and respiratory failure account for approximately 40% of PICU hospitalizations and have a 7–15% death rate [3, 4]. Those who survive acute respiratory infections in the PICU may experience lasting effects. Pollution exposure can harm the pulmonary system, increasing the risk of continual damage and the development of diseases including COPD, bronchitis, and asthma. The danger is significantly increased by surroundings such as exposure to tobacco smoke, air pollution, and interior contaminants. Given that undernourished children are more susceptible to respiratory illness, hunger is a further threat [5]. Machine learning-based treatment has the potential to increase accuracy, help consumers, and reduce physicians' efforts [6]. Paediatric respiratory illnesses can be accurately diagnosed using machine learning models like Gradient Boosting, Random Forest, Decision Tree, CatBoost, K-Nearest Neighbours and AdaBoost. The Random Forest Classifier model demonstrated its dependability in assisting with clinical choice making for paediatric patients by achieving the best accuracy of 99.56% across the evaluated algorithms. The contributions of this paper are given below:

- Random Forest feature selection, Min-Max scaling, and Random Oversampling for balanced data are examples of preprocessing.
- Random Forest, CatBoost, Gradient Boost, Decision Tree, KNN, and AdaBoost are examples of integrated machine learning models.
- Hard and soft voting ensembles have been utilized to improve selection as well as stability.
- SHAP and LIME were applied to the Random Forest model to provide both global and local interpretability.
- Enhanced medical tracking, decreased diagnostic complexities, and increased clinical effectiveness.
- The best accuracy along with AUC were attained by Random Forest.

II. Literature Review

Machine learning has been investigated in a number of research for the prediction and treatment of paediatric respiratory disorders. Using intensive care units, Liu et al. [7] developed a model which forecasts juvenile respiratory arrest and demonstrated promise with an accuracy of 88.9% and an AUC of 87.1%. When Spathis et al. [8] used Random Forest on clinical data from Thessaloniki, Greece, they were able to diagnose persistent lung conditions with 97.7% accuracy. In order to estimate the extent of bronchiolitis, Wang et al. [9] used ensemble learning with XGBoost and Gradient Boosting to analyse wearable sensory information. Aykanat et al. [10] used Gaussian Naive Bayes to classify respiratory illnesses with 98% accuracy.

Asthma and infection related prediction have also been studied, since these outcomes often affect treatment selection and follow up planning. Chang et al. [11] used machine

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learning to predict childhood asthma in the USA, and Random Forest achieved 90.9% accuracy. XGBoost with SHAP was used by Allgaier et al. [12] to predict germs in paediatric respiratory infections with 95% accuracy, where the inclusion of SHAP supports clinician facing explanation of influential factors rather than only a final label. For paediatric pneumonia, Huang et al. [13] used convolutional neural networks (CNNs) on chest X rays to develop an AI based predictive method that showed clinical reliability, reflecting the growing role of deep learning in imaging supported diagnosis. A combined method of learning that combines XGBoost, Random Forest, and Neural Networks was created by Li et al. [14] in order to enhance the detection of paediatric respiratory diseases, suggesting that complementary learners can reduce weaknesses of individual models. Lastly, after applying many combined models to a PICU database, Prithula et al. [15] found that CatBoost had an accuracy of 89.32% and an AUC of 85.2%, reinforcing that boosting based approaches remain competitive for complex clinical data distributions.

III. Materials And Methods

A system for formal machine learning was created to forecast respiratory illnesses in children (Fig 1). Min-Max scaling, mean imputation, label encoding, and column cleaning were part of the data preprocessing. Class imbalance was resolved using Random Oversampling, and key features were chosen based on Random Forest significance. Single classifiers and voting ensembles were used.

A. Dataset

To address respiratory infections, a dataset consisting of 801 children's biological, clinical and paraclinical was taken from Children's Hospital of Rabat, Morocco. This dataset contains 12 numerical and 74 nominal data of children's symptoms and illness conditions. Here, the age of the children are under 6 years. The decision column predicts the condition of illness of that children. Based on these features, the aim of our research is to establish a model that can accurately predict the condition of that child.

B. Data Preprocessing

Several preprocessing techniques were applied to make the dataset clean and usable for our model. One of the important steps of preprocessing is to address the missing values in our dataset. We used mode to find the missing values for categorical data in the dataset, and mean was used to find the missing values for numerical values. For normalization, we have used the Min-Max scaler to set the values in a standard range. We have used label encoding for categorical values. Random Oversampling technique was used to fix the class imbalance in our dataset. For feature selection, we have used the Random Forest importance score and selected the important features for our model.

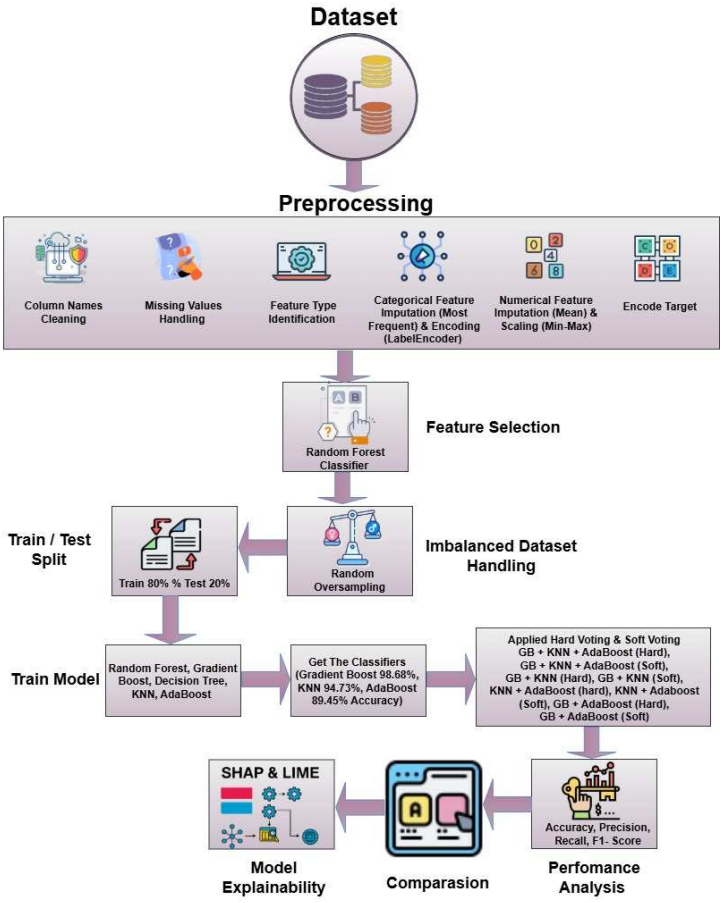


Fig 1: Working Procedure of Proposed Pediatric Respiratory Disease Prediction System

C. Feature Selection and Importance

The top 20 features for pediatric respiratory disease classification are shown in Fig. 2. Important features are given as follows: Detection of DNA and RNA (Alleles 3 and 4), admission date, and household size, which relate to pathogen type, seasonality, and transmission risk. Clinical measures, including procalcitonin, C-reactive protein, respiratory rate, and oxygen saturation (SaO_2), indicate severity and respiratory status. Haemoglobin, iron, and ferritin further reflect oxygen transport and inflammation. These features are helpful to increase the performance of our model. Emphasizing the aforementioned characteristics improves the predictive performance and reduces the impact of extraneous covariates. Within the context of the current investigation, repetition of a Random Forest classifier [17] was chosen as the basic classifier, due to its robustness and its ability to measure feature importance, easing interpretation of which features from the deep layers of the network inform the final decision. Contamination is represented by the Pij term, where the efficacy of a split at node j is described by a distribution relationship between impurity and split that is described by Eq. 1.

$$\hat{y} = \arg\max_{c \in C} \frac{1}{T} \sum_{t=1}^{T} I(h_t(x) = c) \quad \textbf{(1)}$$

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Here, the final predicted class is ŷ, the total number of decision trees is T, h_t(x) is the prediction of the t-th tree for input x, C is the set of possible classes, and I(.) is the indicator function that counts votes for each class.

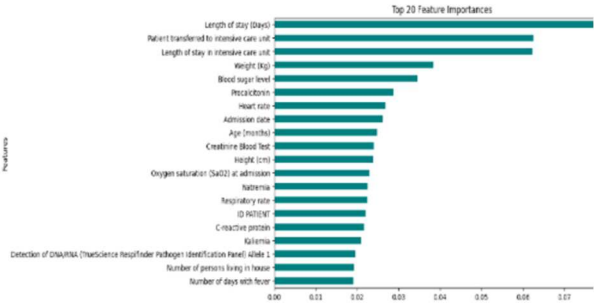


Fig 2. Feature Importance Score

D. Machine Learning Models

For classification performance improvement, we have used several supervised machine learning models and split the dataset where 80% was for training, and 20% was for testing. It was done so that the model can be tested with the data that were not used for training. We have used 6 machine learning models, which are: Random Forest, CatBoost, Gradient Boosting, Decision Tree, KNN, and AdaBoost. Ensemble tree methods capture nonlinear and mixed feature interactions well but KNN provides a distance based baseline. Additionally, we used a hard-voting ensemble combining Gradient Boosting, KNN, and AdaBoost to determine if different decision patterns are more stable. Overall, this workflow allows the comparison of individual models and the assessment of a combined learning strategy.

IV. Result Analysis

A. Environment Setup

The dataset was first divided into training and testing subsets using randomization to prevent data leakage and inflated performance. As shown in Table 1, experiments were conducted under controlled computational conditions. All models were implemented in Python using Google Colab on a notebook environment with 71 GB disk space, a 2 core Xeon 2.2 GHz CPU, a Tesla K80 GPU, and 12.68 GB RAM.

B. Performance Metrics

The performance matrices we have used for the pediatric respiratory diseases dataset model evaluation are Accuracy, Precision, Recall, and F1 score. These are obtained from a confusion matrix, where □ and □ are correct and incorrect outcomes, respectively, and □ and □ represent positive and negative classes. Here, □□ and □□ are correctly classified cases, while □□ and □□ are wrong classifications. Accuracy is shown in Eq. (2), precision in Eq. (3), recall in Eq. (4), and F1 score in Eq. (5) as follows.

$$Accuracy = \frac{TP+TN}{TP+FP+TN+FN} \quad \textbf{(2)}$$

$$\Pr ecision = \frac{TP}{TP+FP} \quad \textbf{(3)}$$

$$Recall = \frac{TP}{TP+FN} \quad \textbf{(4)}$$

$$F1\ Score = \frac{2\times(\Pr ecision\times Recall)}{\Pr ecision+Recall} \quad \textbf{(5)}$$

C. Result Analysis

From Fig. 3, we can see that the Random Forest classifier achieved the highest accuracy of 99.56%. Other classifiers performed well, which are given as follows: CatBoost achieved 98.90%, Gradient Boost achieved 98.68%, and Decision Tree achieved 98.68%, KNN achieved 94.73%, and AdaBoost achieved the lowest accuracy of 89.45%.

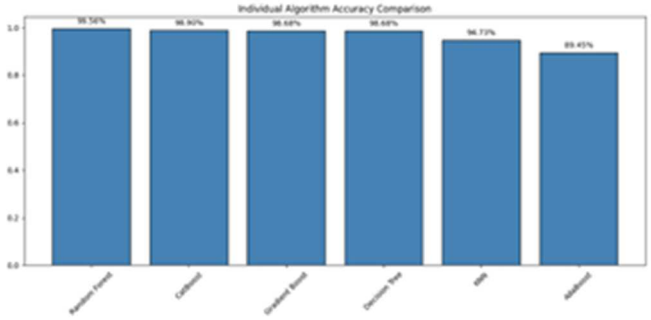


Fig 3: Performance comparison of machine learning models

Table I. shows the performance of 6 machine learning classifiers. Here, we can observe that the Random Forest classifier performed the best, achieving 99.56% accuracy, precision, recall, and F1 score. AdaBoost achieved the poor performance of accuracy of 89.45%..

TABLE I. Performance Metrics For Individual Algorithms

Models	***Accuracy***	***Precision***	***Recall***	***F1- Score***
Random Forest	99.56 %	99.56 %	99.56 %	99.56 %
CatBoost	98.90 %	98.92 %	98.90 %	98.89 %
Gradient Boost	98.68 %	98.71 %	98.68 %	98.67 %
Decision Tree	98.68 %	98.72 %	98.68 %	98.67 %
KNN	94.73 %	95.12 %	94.72 %	94.61 %
AdaBoost	89.45 %	89.88 %	89. 45 %	89.49 %

For ensemble learning, we have used GB, KNN, and AdaBoost. We used both soft and hard voting techniques for our evaluation. Table II. shows that, in hard voting , GB+KNN achieved the highest accuracy, among the other ensemble models, of 99.34%, KNN+AdaBoost achieved an accuracy of 92.30%, and GB+KNN+AdaBoost achieved 98.68%. In soft voting, GB+AdaBoost performed better, achieving an accuracy of 98.68%. Thus, we can observe that hard voting performed better than soft voting. So, hard voting can be taken into consideration for this dataset.

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TABLE II. PERFORMANCE METRICS FOR SOFT & HARD VOTING ALGORITHMS

Models	***Accuracy***	***Precision***	***Recall***	***F1-Score***
GB+AdaBoost (Soft)	98.68%	98.71%	98.68 %	98.67 %
GB+KNN (Soft)	97.58%	97.67%	97.58 %	97.56 %
KNN+AdaBoost (Soft)	94.72%	95.12%	94.72 %	94.61 %
GB+KNN+AdaBoost (Soft)	98.46%	98.49%	98.46 %	98.45 %
GB+AdaBoost (Hard)	91.86%	92.68%	91.86 %	91.92 %
GB+KNN (Hard)	99.34%	99.34%	99.34 %	99.33 %
KNN+AdaBoost (Hard)	92.30%	93.22%	92.30 %	92.36 %
GB+KNN+AdaBoost (Hard)	98.68%	98.71%	98.68 %	98.67 %

IV. Confusion Matrix

From Fig. 4, we can observe that the confusion matrix of Random Forest has the best performance. This means the classification has been perfect for each testing case. Again, AdaBoost has poor performance, which can also be observed from its confusion matrix. Therefore, we can say that Random Forest has been the best option for this classification, and AdaBoost is not a suitable option for this classification.

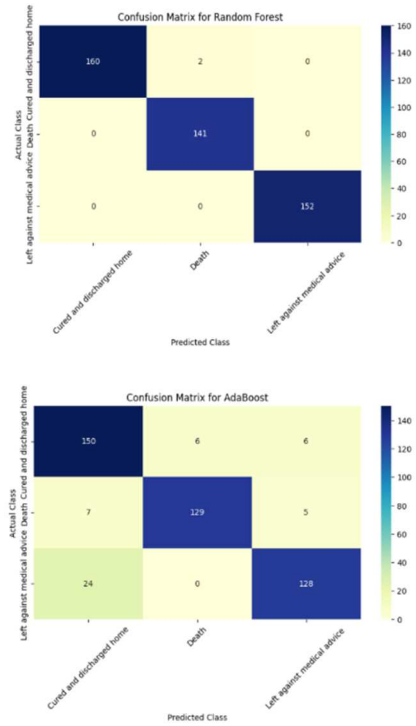


Fig 4: Comparison of Confusion Matrices: Best (Random Forest) and Worst (AdaBoost) Performing Models

E. ROC Curve

The ROC curves of single classifiers and ensemble models are shown in Fig. 5. An AUC score of 1.00 was achieved by Gradient Boosting, Random Forest, and CatBoost. Again, we can observe that Decision Tree and KNN achieved 0.99 AUC score. AdaBoost has an AUC score of 0.96. As for the ensemble models, hard voting with GB + KNN + AdaBoost achieved an AUC score of 0.99. Soft voting of GB + KNN, GB + AdaBoost, and GB + KNN + AdaBoost performed well, achieving an AUC score of 1.00. In this case, soft voting performed better than hard voting.

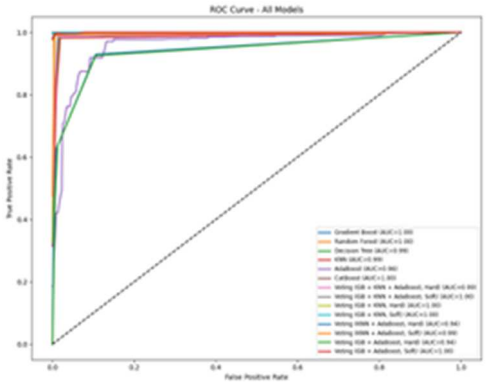


Fig 5: Performance Evaluation: ROC Analysis of Multiple Classifiers.

F. Model Interpretation

For model transparency, we have used the SHAP (SHapley Additive exPlanations) [18] summary plot for our best case, the Random Forest model. It was done to improve interpretability, which is shown in Fig. 6. The strongest

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features can be observed here, are Length of stay, ICU duration, and ICU transfer. It indicates that hospitalization severity and prognosis risk. Clinical features, including heart rate, respiration rate, oxygen saturation, CRP, procalcitonin, age, and weight, were strongly linked to the outcomes. Features like Blood glucose, creatinine, natremia, and kaliemia showed standard interpretability. From SHAP analysis, we can observe how the model separates recovery from illness in pediatric respiratory diseases.

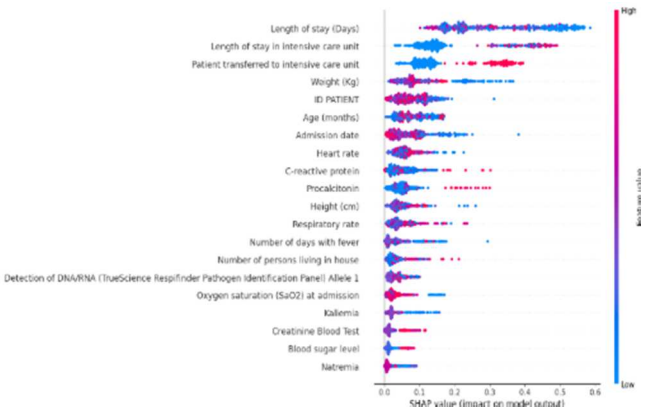


Fig 6: SHAP Summary Plot for Global Feature Importance in Pediatric Respiratory Prediction

Fig. 7 shows the interpretability of the Random Forest model using LIME [19] for pediatric respiratory classifications. One of the major features was ICU progression. Its absence shows as a key negative factor. Other features that contributed positively are: Age, weight, and biochemical markers such as blood sugar, creatinine, and kaliemia. Decisions van have an impact based on features like Length of stay and admission date. Thus, we can say that case specific explanations are offered by LIME. Therefore, it supports clinical trust and transparency.

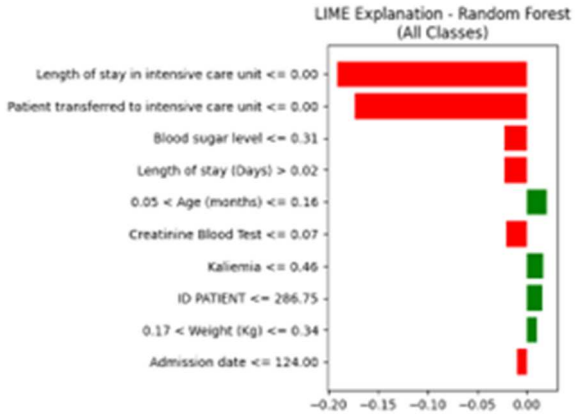


Fig 7: LIME Explanations for Local Interpretability of Pediatric Respiratory Outcomes

The proposed Random Forest model achieved an accuracy of 99.56% that outperforms the existing studies on pediatric respiratory disease prediction. It is shown in Table III. This performance can be validated as we did class balancing, Min-Max scaling and selecting important features. Unlike prior studies, we experimented both single classifiers and ensemble strategies along with explainability. This transparency was done using Explainable AI. Therefore, we can say that Random Forest can be a reliable model for pediatric respiratory diseases prediction.

TABLE III. COMPARING SEVERAL MACHINE LEARNING ALGORITHMS WITH OUR PROPOSED MODEL

References	*Data Split*	*Best Model*	*Accuracy*	*XAI Methods*
C. F. Tso[et al 20]	80-20 TRAIN-TEST	Extreme Gradient Boosting	91.9 %	SHAP
C. Y. Effah et al [21]	80-20 TRAIN-TEST	Random Forest	97 %	None
D. M. Kothalawala et al [22]	60-33 TRAIN-TEST	Support Vector Machine	94.1 %	SHAP
R. Kumar et al [23]	70-30 TRAIN-TEST	Random Forest	96 %	SHAP, ELI5, QLattice, LIME
Proposed Model	80-20 TRAIN-TEST	Random Forest	99.56 %	SHAP & LIME

V. CONCLUSION

Respiratory diseases are a serious concern for children's health. Early detection is very important, along with its accurate results. This is needed to provide early assistance and the possibility of early recovery. For a transparent diagnosis, we have presented a machine learning workflow. Initially, we have preprocessed the dataset by finding missing values, label encoding, and min-max scaling. Then we addressed the class imbalance and fixed it using Random Oversampling. Then we selected the important features by analyzing the important scores obtained using Random Forest importance. Then we moved to model training. Before doing that, we split the dataset into 80% train and 20% test, so that the model can be checked and validated. Then we used 6 different single machine learning classifiers and noted their evaluation. Then we used ensemble strategies where we showed both soft and hard voting performance. We observed that Random Forest showed the best accuracy of 99.56%. Finally, model interpretability was shown using SHAP and LIME, which supports its transparency. Future work will extend the study through broader datasets, cross-validation, and real time monitoring integration. Also upcoming studies will optimize preprocessing techniques. This work expands the use of machine learning in paediatric respiratory disease prediction.

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